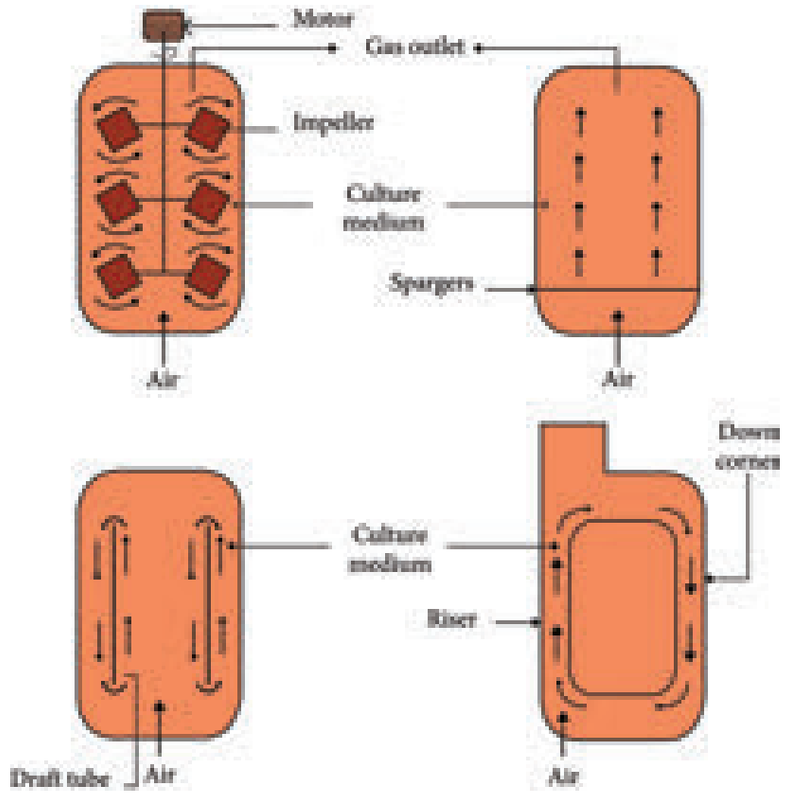


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Figure 6.6: Components of Fermentor

Table 6.2: Components and their uses:

S. no	Parts of fermenter	Functions
1	Impeller (agitator)	To stir the media continuously and hence prevent cells from setting down and distribute oxygen throughout the medium. Impellor speed decreases as the size of the fermenter increases
2	Sparger (aerator)	Introduce sterile oxygen to the media in case of aerobic fermentation process
3	Baffles (vortex breaker)	Disrupt vortex and provide better mixing
4	Inlet Air filter	Filter air before it enter the fermenter
5	Exhaust Air filter	Trap and prevent contaminants from escaping
6	Rota meter	Measure flow rate of Air or liquid
7	Pressure gauge	Measure pressure inside the fermenter
8	Temperature probe	Measure and monitor change in temperature of the medium during the process
9	Cooling jacket	To maintain the temperature of the medium throughout the process
10	pH probe	Measure and monitor pH of the medium
11	Dissolve oxygen probe	Measure dissolve oxygen in the fermenter
12	Level probe	Measure the level of medium
13	Foam probe	Detect the presence of the foam
14	Sampling point	To obtain samples during the process
15	Valves	Regulation and control the flow liquids and gases

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